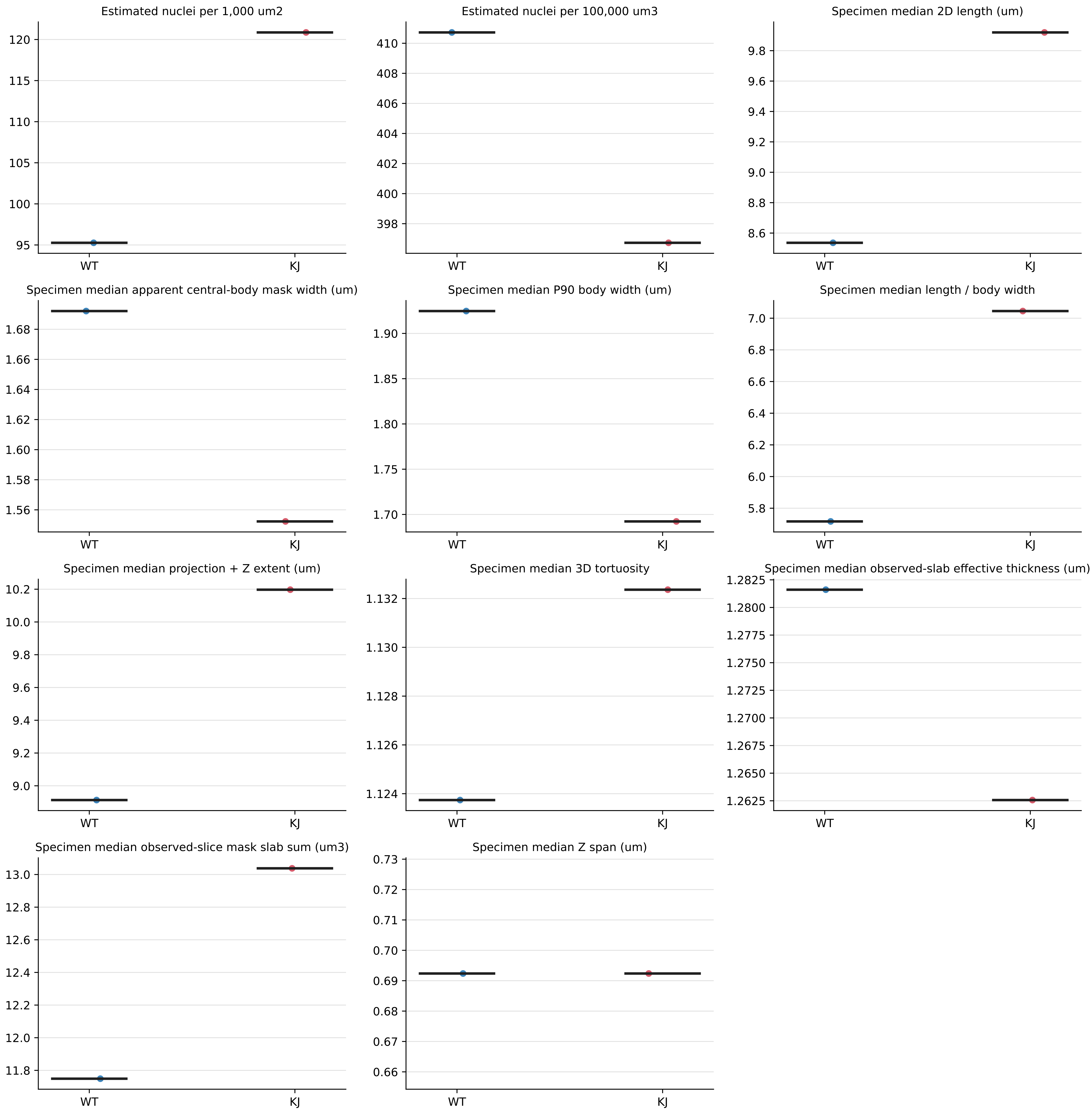
Specimen-level group comparison
Each point is one biological specimen; bars show medians



How to read the specimen-level comparison

Groups shown: WT (reference) and KJ (comparison). Each colored point is one biological specimen. The black horizontal line is the group median. Individual nuclei are not treated as independent biological replicates.

COUNT AND NORMALIZATION PANELS

Primary count
N = number of technical-valid 3D tracks. Morphology warnings remain included; obvious leakage, invalid geometry, and unresolved multi-object components are excluded.

ROI area
A = ROI pixels x (XY calibration in um/pixel)^2

Sampled depth and volume
D = included Z slices x Z step (um)
V = A x D

Estimated nuclei per 1,000 um2
(N / A) x 1,000
An area-normalized projection-style density. It does not correct for different stack depths.

Estimated nuclei per 100,000 um3
(N / V) x 100,000
A sampled-volume density. This is preferable when stack depths differ, but it can still be influenced by acquisition depth, stack boundaries, and cross-slice tracking.

MORPHOLOGY PANELS

Specimen median 2D length
For each reconstructed nucleus, take its maximum calibrated centerline length across observed slices; then take the specimen median.

Specimen median 2D width
Median calibrated mask width for each nucleus, summarized by the specimen median.

Length / width
Centerline length divided by mask width. Larger values indicate a more elongated, slender object.

Projection + Z extent
sqrt(maximum lateral 2D length^2 + Z span^2). This is a calibrated projection-plus-Z estimate, not a surface-mesh length.

3D tortuosity
Estimated 3D path length / 3D end-to-end distance. A value near 1 is straight; larger values are more curved.

Volume
Sum of filled-mask pixels across slices x XY pixel area x Z step.

Effective thickness
2 x sqrt((observed-slice slab sum / projection + Z extent) / pi).
This is a diameter proxy and is PSF- and segmentation-sensitive.

Z span
(last Z index - first Z index) x Z step.

INTERPRETATION: Use specimen-level medians and effect sizes for reference-versus-comparison analysis. A morphology warning annotates an unusual but measurable nucleus and does not remove it. Count-density results should be checked for stack-depth sensitivity. Volume and effective thickness should be compared only between matched microscope settings. Statistical results are exploratory and are reported in specimen_group_comparisons.csv.

What each measurement means biologically

Every panel summarizes one value per specimen. The question below each heading describes what that panel can help answer; it does not by itself prove a biological mechanism.

Estimated nuclei per 1,000 um2
Question: How many nuclei are present per unit of sampled 2D ROI area?
Meaning: An XY-area density. Higher values mean more reconstructed nuclei within the sampled ROI footprint, but deeper stacks can increase this number because stack depth is not included.

Estimated nuclei per 100,000 um3
Question: How many nuclei are present per unit of sampled 3D volume?
Meaning: A volume-normalized density. Higher values indicate more nuclei within a sampling denominator made by repeating the same 2D ROI through the nominal stack depth. It is not anatomical tissue, seminal-vesicle, or whole-organ volume.

Specimen median 2D length
Question: Are the nuclei typically longer in the image plane?
Meaning: The typical maximum visible centerline length for a nucleus across its observed slices. Higher values indicate longer projected nuclei; the specimen median limits the influence of extremes.

Specimen median 2D width
Question: Are nuclei typically broader or thinner?
Meaning: The typical calibrated width of the filled nucleus masks. Higher values indicate broader nuclei. Width is influenced by optical resolution, PSF, focus, and mask boundaries.

Specimen median 2D length / width
Question: Are nuclei more elongated or more rounded?
Meaning: A shape ratio. Higher values indicate long, slender nuclei; lower values indicate shorter, broader, or more rounded nuclei. It should be interpreted together with length and width.

Specimen median projection + Z extent
Question: Are nuclei longer after accounting for their Z orientation?
Meaning: Combines maximum lateral length with calibrated Z span. It can exceed 2D length when a nucleus extends through several planes. It is a projection-plus-Z estimate, not a surface trace.

Specimen median 3D tortuosity
Question: Are nuclei straighter or more curved?
Meaning: The calibrated path through ordered slice centroids divided by its end-to-end distance. A value near 1 indicates a straight cross-slice trajectory; it does not trace the full nuclear surface.

Specimen median effective thickness
Question: Is the reconstructed nucleus effectively thicker?
Meaning: A diameter proxy calculated from volume divided by length. Higher values suggest thicker objects, but this is not a direct width measurement and is PSF- and segmentation-sensitive.

Specimen median volume
Question: How much calibrated 3D mask volume does a typical nucleus occupy?
Meaning: Filled-mask area accumulated over observed Z slices only; missing slabs are not interpolated. Higher values can reflect longer or thicker nuclei, but volume also depends on mask thresholds, Z sampling, and optical resolution.

Specimen median Z span
Question: Through how much physical optical depth is a nucleus observed?
Meaning: Distance from its first to last linked Z plane. Larger values indicate greater axial extent or tilt. A single-slice track has zero endpoint-to-endpoint Z span and is not automatically invalid.

Use the direction and size of the specimen-level shift together with confidence intervals, effect sizes, acquisition checks, and representative overlays. Do not choose a biological conclusion from a p-value or one panel alone.